

Use-Case Flow Specification

Food Bank Surplus Redistribution Platform | Problem Statement #62 | Lab 1 – PES University

Use Case	UC-03: Reserve Pickup Window
Primary Actor	Shelter Coordinator
Secondary Actor	Donor Partner (notified of the reservation)
Includes	UC-06: Validate Reservation Eligibility
Related Requirements	FR-003, FR-004, NFR-002
Trigger	Coordinator chooses “Reserve” on a food batch and selects a pickup window.

Preconditions

1. Shelter Coordinator is registered and logged in.
2. The shelter account is verified.
3. The food batch exists, is Available and has not expired.

Postconditions (on success)

1. A reservation record is created and the batch status changes to Reserved.
2. The chosen pickup window and pickup details are stored.
3. A unique pickup verification code is generated and linked to this reservation.
4. The Shelter Coordinator receives confirmation; the Donor Partner is notified.

Main Success Scenario

1. Shelter Coordinator logs in.
2. System displays the available food batches (UC-02).
3. Coordinator selects a food batch.
4. System shows quantity, expiry countdown, dietary tags, pickup location and pickup windows.
5. Coordinator selects an available pickup window and requests the reservation.
6. System runs Validate Reservation Eligibility (UC-06): shelter is verified, batch is Available, batch is not expired, window is valid.
7. System creates the reservation and updates the batch status to Reserved.
8. System generates the unique pickup verification code.
9. System shows the confirmation, code and pickup details to the Coordinator and notifies the Donor Partner.

Alternate Flow A1 – Batch no longer available (at step 6)

1. Another shelter reserved the same batch before this request was confirmed.
2. System finds the batch is no longer Available during eligibility validation.
3. System rejects the request; no reservation is created.
4. System shows “This batch has already been reserved.”
5. Coordinator returns to the available food board (use case ends).

Alternate Flow A2 – Batch expired or shelter not verified (at step 6)

1. System detects that the batch has expired, or that the shelter is not verified.
2. System rejects the request and shows the reason (“Food expired” or “Shelter not verified”).
3. Coordinator returns to the food board (use case ends).

Business Rules

Only verified shelters may reserve. Expired food cannot be reserved. Availability is checked before confirming. One batch cannot have two reservations. Each pickup code is unique to one reservation.